2. Minimum Jerk

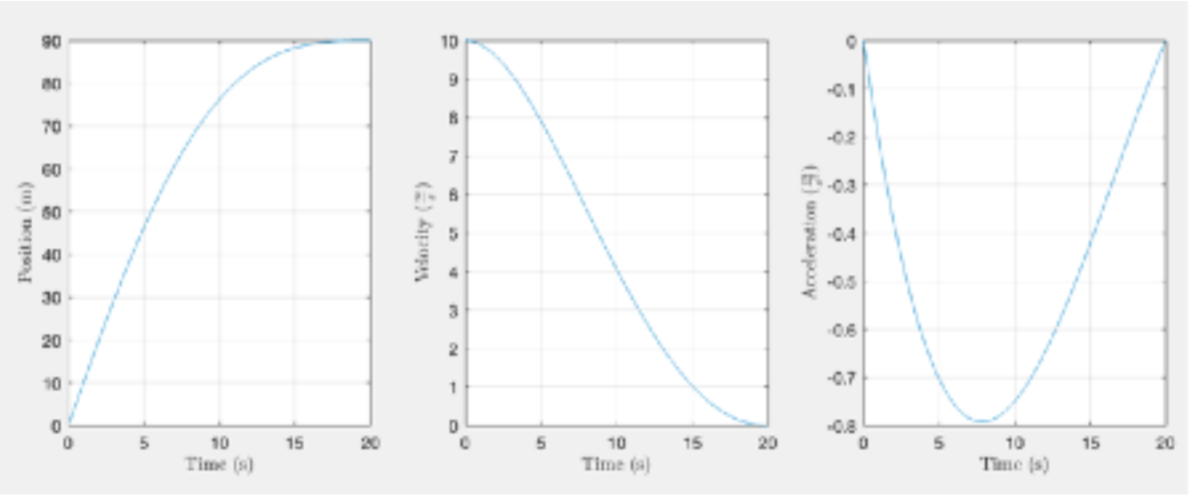
Optimal Control Problem:
Square jerk function is the Lagrangian cost of our OCP → solved using *MATLAB* imposing *initial* and *final conditions* → obtaining **optimal solutions** and extracting **opt**. **coefficients**.

$$s_{\text{opt}}(t) = c_1 t + \frac{1}{2}c_2 t^2 + \frac{1}{6}c_3 t^3 + \frac{1}{24}c_4 t^4 + \frac{1}{120}c_5 t^5$$

$$v_{\text{opt}}(t) = c_1 + c_2 t + \frac{1}{2}c_3 t^2 + \frac{1}{6}c_4 t^3 + \frac{1}{24}c_5 t^4$$

$$a_{\text{opt}}(t) = c_2 + c_3 t + \frac{1}{2}c_4 t^2 + \frac{1}{6}c_5 t^3$$

$$j_{\text{opt}}(t) = c_3 + c_4 t + \frac{1}{2}c_5 t^2$$



Subsequently, the application *MATLAB coder* has been used to generate the **C** code needed to implement the **agent**.

3. Stopping Primitive

Define a primitive function to **STOP** the vehicle within selected **boundaries**.

- **Final** Conditions:
  $a_f = 0$ $\rightarrow$ final acceleration equal to zero;
  $v_f = 0$ $\rightarrow$ final velocity equal to zero;
  $s_f = \bar{s}_f$ $\rightarrow$ final position equal to the position of the traffic light;
- **Initial** Conditions:
  $v_0 = \bar{v}_0 \rightarrow$ initial velocity given by the sensor;
  $a_0 = \bar{a}_0 \rightarrow$ initial acceleration given by the sensor;

Find the optimal **time** solution:
the total cost is derived w.r.t. t_f and imposed equal to zero $\rightarrow$ new function to define the optimal **time** needed to reach a given **final position** (traffic light).